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All-pass Si_3N_4 metasurface filter for advanced photonic applications: metalenses, vortex beams, and holography

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Abstract

Dielectric metasurfaces have emerged as a powerful platform for advanced photonic applications, offering unparalleled control over light manipulation. In this study, we design and analyze an all-pass silicon nitride (Si_3N_4) metasurface filter composed of nanopillars to investigate the efficiency and performance of the resulting metalenses, optical vortex beams, and holographic imaging systems operating in the near-infrared (NIR) and mid-infrared (MIR) regions. Starting with the metalenses, we examine how their numerical aperture (NA) influences their focusing performance. Our results show that for both 1100 nm and 8000 nm excitation wavelengths, the metalenses achieve a focusing efficiency of 85% at a NA of 0.707, and this efficiency increases to approximately 90% as the NA is reduced to 0.447. Next, we integrate lens and spiral phase plate functionalities into a single Si_3N_4 metasurface to generate focusing optical vortex (FOV) beams. At an operating wavelength of 1100 nm and a topological charge of $l = 5$, the generated FOV beam exhibits high purity with a focusing efficiency exceeding 90%. These performance enhancements are attributed to the metasurface's near-unity transmission across a broad wavelength spectrum. Finally, after validating the computer-generated hologram phases experimentally using a spatial light modulator, which can be accurately reproduced in optical systems, we employ Si_3N_4 metasurfaces for transmissive holographic imaging. Their high transmission efficiency enables successful far-field reconstruction of holographic images at excitation wavelengths in the NIR and MIR regimes, with image-generation efficiency reaching up to 80%. Our results firmly establish and highlight the potential of Si_3N_4 metasurfaces as a useful platform for advancing photonic systems, offering exceptional efficiency and functional versatility across diverse applications.

1. Introduction

The manipulation of light at the nanoscale has revolutionized photonic technologies, enabling the development of compact and highly efficient devices. By engineering flat structures known as metasurfaces, which comprise tiny meta-atoms capable of manipulating light in unprecedented ways, researchers can transcend the limitations of natural materials. These metasurfaces dramatically enhance light-matter interactions, achieving precise control over electromagnetic wave properties such as phase, amplitude, and polarization of light at nanoscale dimensions [1, 2]. One of the most compelling features of metasurfaces is their ability to manipulate electromagnetic waves at subwavelength scales, a capability unattainable with traditional optical components [3, 4]. By precisely tailoring the geometry, orientation, and arrangement of these nano-scatterers, metasurfaces can implement complex optical functions such as beam steering, wavefront shaping, and holography within an ultrathin layer [5, 6].

For instance, metasurfaces have been instrumental in the development of metalenses with diffraction-limited focusing capabilities, capable of correcting optical aberrations and achieving high numerical apertures (NAs) [7, 8]. Additionally, metasurfaces enable the generation and manipulation of structured light beams, such as optical vortex beams carrying orbital angular momentum (OAM), which are essential for advanced optical communication and quantum information processing [9–11]. Dielectric metasurfaces, in particular, have gained significant attention due to their low absorption losses and high transmission efficiencies, making them crucial for practical applications across a wide spectral range [12, 13]. For instance, conventional plasmonic metasurfaces often suffer from high ohmic losses due to their metallic constituents, which limits their efficiency [3]. Additionally, even some dielectric metasurfaces such as silicon (Si) exhibit a significant absorption loss in the visible and near-infrared (NIR) [14]. For instance, Chong *et al* demonstrated complex wavefront control using a highly efficient polarization-insensitive holographic metasurface based on resonant silicon meta-atoms. However, since the transmission efficiency of the silicon meta-atoms is as high as 82%, which is much lower than the ideal value, it resulted in a 40% imaging efficiency [15].

To address these limitations, Si_3N_4 has attracted attention as an alternative dielectric material for metasurface applications. Si_3N_4 possesses a moderate refractive index and low absorption in the NIR and mid-infrared (MIR) spectral regimes, as well as compatibility with standard fabrication processes [16]. Additionally, Si_3N_4 metasurfaces can function as nearly all-pass filters, enabling high transmission efficiencies across a broad spectral range. This material can be engineered to provide the desired phase shifts while minimally attenuating the transmitted light [17]. The nearly all-pass behavior allows for the design of metasurfaces that highly manipulate light efficiently, overcoming limitations of significant losses due to material absorption over a broad spectral range. Moreover, such a metasurface, which is capable of controlling both phase and amplitude, opens up a wide range of applications, especially in multimodal imaging, through the precise manipulation of light [18].

Subsequently, in this study, we design and analyze metalenses, optical vortex beams, and holographic imaging systems using the nearly all-pass properties of Si_3N_4 metasurfaces composed of nanopillars. We focus on devices operating in the NIR ($\lambda = 1100$ nm) and MIR ($\lambda = 8000$ nm) spectral regimes. Using rigorous simulation methods, we investigate the performance of these metasurfaces in terms of focusing efficiency, beam purity, and imaging quality. Our results demonstrate that Si_3N_4-based metalenses achieve high focusing efficiencies across various NAs. For example, at a NA of 0.98, the metalens maintain a focusing efficiency greater than 65% at 1100 nm and 8000 nm, significantly outperforming a-Si metalenses, which achieve only 35% efficiency [19]. This demonstrates that Si_3N_4 metalenses exhibit superior performance, highlighting their potential for high-resolution imaging applications.

Beyond metalenses, we explore the generation of high-purity optical vortex beams using Si_3N_4 metasurfaces that combine the functionalities of a lens and a spiral phase plate. At a topological charge of $l = 5$, the focusing efficiency reaches 90.7%, significantly surpassing the efficiencies reported for all-dielectric metasurfaces made from silicon nanoblocks [20]. This enhancement is attributed to the near-unity transmission efficiency of the Si_3N_4 nanopillars, which minimizes losses and aberrations even at higher topological charges. The ability to generate high-quality, focusing optical vortices is essential for applications requiring ring-shaped intensity distributions or precise control over the OAM of light, such as material processing, optical trapping, and quantum communications [21–23].

Furthermore, we employ these Si_3N_4 metasurfaces for transmissive holographic imaging under NIR and MIR excitation. The high transmission efficiency of Si_3N_4 is crucial for reconstructing high-quality holographic images. We successfully reconstruct images at various distances with an efficiency of 79.9%, demonstrating consistent performance and spatial flexibility, which are essential for applications like augmented reality (AR) and optical data storage. Additionally, we implemented reflective holographic imaging using TiO_2 nanopillars on a silver film to further validate our approach. The phase patterns used to reconstruct these holographic images were also validated using a reflective spatial light modulator (SLM), achieving the expected holographic image with good quality. In summary, our work leverages the advantageous properties of Si_3N_4 to design and analyze metasurfaces that exhibit high efficiency and functionality across multiple applications.

2. Design and method

In this research, we design and analyze metalenses, optical vortex beams, and holographic imaging using dielectric metasurfaces. The used metasurfaces are composed of Si_3N_4 nanopillars with $n = 2.05$, and titanium dioxide (TiO_2) nanopillars [24]. We analyze metalenses operating in the NIR spectra regime ($\lambda = 1100$ nm) and MIR ($\lambda = 8000$ nm) using Si_3N_4 nanopillars. The Si_3N_4 nanopillars were also used to generate optical vortex beams and enable transmissive holographic imaging. Therefore, numerical simulations of the metasurfaces, such as transmittance and reflectance, were performed using rigorous coupled-wave analysis (RCWA) and finite-difference time-domain (FDTD) Solutions (Lumerical, Inc.) and PlanOpSim software [25]. Periodic boundary conditions were applied along the x and y axes, with perfectly matched layers (PMLs) applied in the z-direction for the FDTD [26, 27]. A plane-wave source at normal incidence along the z-direction was used as the light source. After the metasurface was designed and optimized, we employed a metalens design using PlanOpSim software, which involved several key steps [25, 28]: First, we develop a library of structures capable of modulating the phase response of 2π radians. Next, we specify the desired wavefront characteristics, including amplitude and phase, to tailor the component's functionality. Finally, we strategically arrange the structures from the library across the surface to approximate the target wavefront (phase) and achieve the desired functionality accurately. As a baseline example, we created a compact meta-lens optimized for 1100 nm and 8–10 μm operating wavelengths. Unlike metasurfaces that exploit a geometric phase, we utilized nanopillar structures to achieve a 2π phase delay. The local periodic approximation was employed due to its ability to simulate the effects of neighboring meta-atoms without requiring full-wave calculations of the entire area simultaneously. Moreover, at a target excitation wavelength of 1100 nm, we designed and investigated different optical vortex beams with various topological charge values using Si_3N_4 nanopillars. Finally, the computer-generated hologram (CGH) phase of the University of New South Wales (UNSW) logo was validated and accurately reproduced experimentally in an optical system using the PLUTO-2.1 Reflective SLM from HOLOEYE Photonics AG. Thereafter, this phase was used to generate transmissive holographic imaging using the all-pass Si_3N_4 metasurface filter.

3. Results and discussion

3.1. Optical property of Si_3N_4 metasurfaces

The optimization of the meta-atom in figure 1 demonstrates the effectiveness of Si_3N_4 nanopillars in manipulating light in a wide range of wavelengths. Figure 1(a) illustrates a square lattice configuration with nanopillars having a constant height of 2 μm, a radius varying from 50 to 230 nm, and a periodicity of 550 nm. This design achieves polarization-insensitive control over light propagation, a key feature for applications requiring uniform optical performance regardless of polarization. This is made possible through symmetric nanopillar designs, such as cylindrical shapes, which ensure equal interaction for both transverse electric (TE) and TM modes (see figure S1 in the supplementary information). The near-identical behavior across both polarizations confirms the polarization-insensitivity of the metasurface. This characteristic is critical for devices like metalenses, which need to manipulate light regardless of its polarization. Such polarization independence is particularly advantageous in systems requiring broadband light manipulation [29]. Additionally, this is important for applications that require precise light control without polarization dependency, allowing easy integration and implementation in other metasurface devices, especially those used in imaging [30].

Figure 1(b) compares different simulation methods, such as Lumerical RCWA, FDTD, and PlanOpSim, demonstrating consistency across all methods and validating the robustness of design optimization. These simulations show that optimized nanopillars efficiently manipulate light and achieve high transmittance across the NIR spectrum [31, 32]. Figures 1(c) and (d) demonstrate that this metasurface design not only ensures high transmission efficiency but also provides precise phase modulation, which is crucial for applications in designing efficient metalenses. Metalenses require a 2π phase shift to focus light effectively. This metasurface's ability to control light with minimal loss across the 1000–1600 nm range makes it ideal for NIR cameras, light detection and ranging systems, and adaptive optics in fields such as autonomous vehicles and aerial imaging, where high resolution and broadband functionality are necessary [7].

Moreover, the all-pass filter behavior of the metasurface, highlighted in figures 1(b) and (c), demonstrates a transmission efficiency exceeding 98% in the 1000–1600 nm range, a critical factor for many optical applications. The polarization-insensitive design and precise phase control ensure effective light manipulation, key for imaging. The metasurfaces' high efficiency and broadband response make them suitable for high-performance optical devices requiring both effective light manipulation and broad wavelength coverage. These metasurfaces also have great potential in telecommunications, as the 1000–1600 nm range aligns with the key wavelengths used in fiber optic communications. Their high transmission efficiency minimizes signal loss, enhancing data transmission rates and improving system performance. Integrating these metasurfaces into optical networks makes it possible to increase fiber optic throughput and optimize infrastructure for high-speed internet and telecommunications.

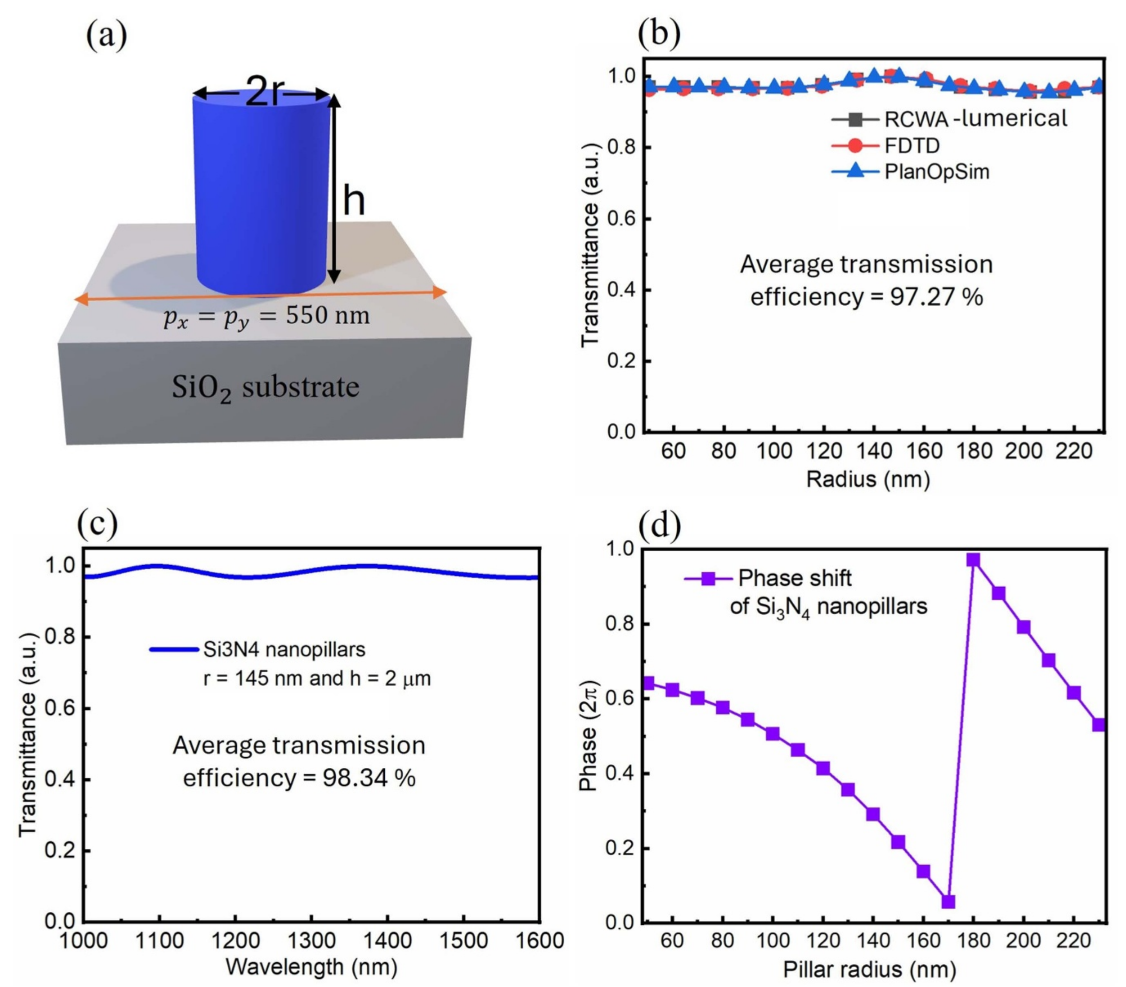

Figure 1. Meta-atom optimization. (a) Schematic illustration of Si_3N_4 nanopillars arranged in a square lattice as meta-atoms, with the height $h = 2$ μm, radius r varying from 50 to 230 nm, and a period of 550 nm as the design parameters. This configuration is optimized to control light propagation through the metasurface. The demonstrated nanopillars exhibit consistent transmission efficiency for both TM and TE modes (see figure S1 in the supplementary information). (b) A comparison of transmission efficiency across different simulation methods (Lumerical RCWA, FDTD, and PlanOpSim) demonstrates the accuracy and reliability of the design simulations. (c) The effect of varying the excitation wavelength on transmission efficiency, and (d) the phase shift induced by the metasurfaces. The period, height, and radius of the nanopillars were systematically varied to determine the optimal set of nanostructures for metalens design.

3.2. NIR metalens

After confirming the nearly ideal transmission efficiency of the Si_3N_4 metasurface and their ability to provide 2π phase shift (see figure 1), we designed the phase profile of the metalens in the NIR regime as follows [1, 33]:

$$\phi(x,y) = \frac{2\pi}{\lambda}\left(\sqrt{x^2+y^2+f^2}-f\right), \tag{1}$$

where, $\phi(x,y)$ represents the phase shift at point (x,y) on the metalens, λ is the wavelength of incident light, and f is the focal length. Additionally, the NA of the metalens is determined by $\mathrm{NA} = n\sin(\theta)$, where $n = 1$ for air and θ is the maximum incident angle of light, calculated as $\theta = \tan^{-1}(d/2f)$, with d being the aperture size (see figure 2(a)) [34]. The Si_3N_4 nanopillars play a key role in sub-wavelength light manipulation for precise focusing at the focal plane. They are designed to deliver a full 2π phase shift, which is crucial for efficient light focusing. The precise arrangement of these nanopillars ensures sharp focusing, and their ability to control phase allows the lens to achieve high resolution, especially at higher NAs. The phase modulation capacity enables the metalens to maintain performance as they operate across different wavelengths and conditions.

Figures 2(b)–(f) highlight how the NA enhances the metalens' ability to focus light. As NA increases from 0.242 to 0.98, the focusing efficiency decreases from 89.8% to 65.0%. This inverse relationship shows that while higher NA values improve resolution, they reduce efficiency. As shown in figure 2(f), the designed high NA metalens (N.A = 0.98) achieves precise light focusing with a focus efficiency

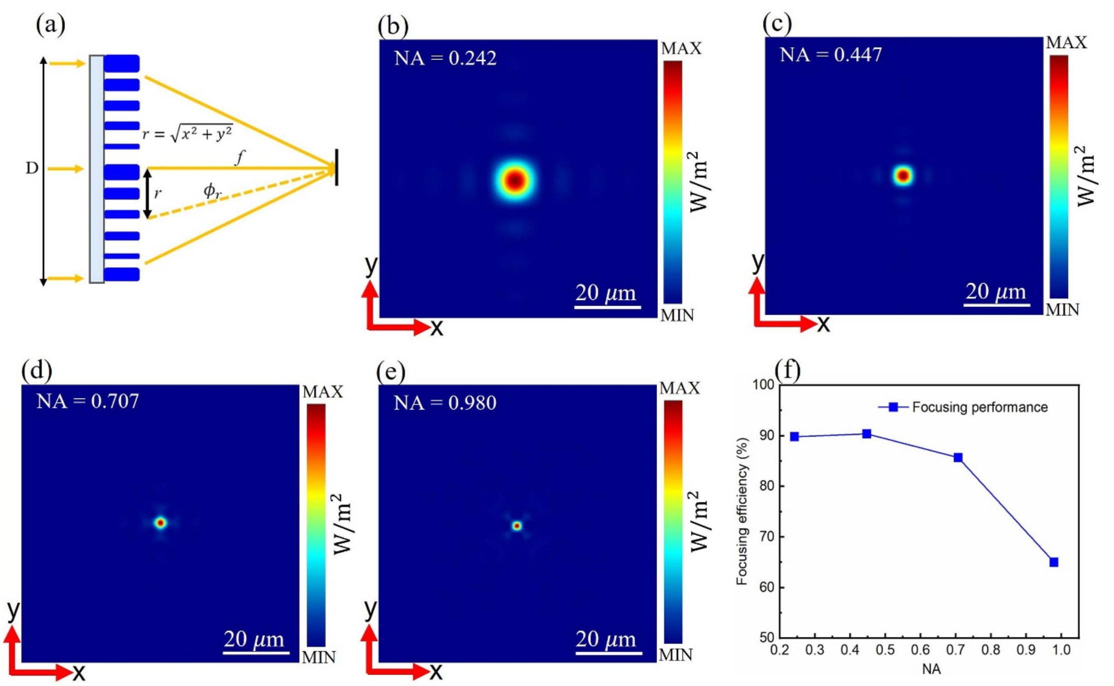

Figure 2. Focusing efficiency of the metalens at different NAs (a) A graphical illustration depicting the working principles of a metalens composed of Si_3N_4 nanopillars. This figure showcases the arrangement and design of the nanopillars, which are engineered to focus incoming light precisely at the focal plane. (b)–(e) The resulting intensity profiles generated at the focal plane for different NAs: (b) 0.242, (c) 0.447, (d) 0.707, and (e) 0.980. (f) A corresponding graph shows the metalens' calculated focusing efficiency as a function of different numerical apertures.

exceeding 65.0%. Here, the focusing efficiency (as computed directly in PlanOpSim) is defined as the fraction of the incident optical power on the metalens surface that is concentrated into a circular aperture in the focal plane with a diameter equal to 3 times the full width at half maximum (FWHM) of the focal spot intensity profile [35, 36]. The reported focusing efficiency is high compared with other nanostrcutred lenses [19, 25]. The focusing efficiency typically decreases at higher NAs because higher NA lenses require rapidly varying phase profiles, especially near the edges. If the spatial sampling (e.g. Si_3N_4 nanopillar spacing) is insufficient to capture these variations accurately, phase errors occur. This results in inefficient focus, as not all light is directed toward the desired focal spot [35, 36]. The intensity profiles for NAs ranging from 0.242 to 0.980 show that the metalens progressively achieve tighter focus with increasing NA (see figures 2(b)–(e)). At lower NAs, the focal spot is broader, indicating lower resolution, but as the NA increases to 0.447 and 0.707, the light becomes more focused. By NA of 0.980, the focal spot is at its sharpest, showcasing the high resolution achievable at larger NAs and securing a higher focusing efficiency.

3.3. MIR metalens

The development of MIR metalenses using Si_3N_4 nanopillars represents a significant advancement in optical device engineering. Si_3N_4 is an excellent material for MIR applications due to its low absorption losses in this spectral region and compatibility with standard fabrication techniques [16]. In our design, we achieved a full 2π phase shift using shorter nanopillars at target wavelengths of 8 μm, despite Si_3N_4 having a lower refractive index compared to silicon. This contrasts with the 25.8 μm height required for all-silicon metasurfaces at 10.6 μm [17], highlighting the efficiency of Si_3N_4 in phase manipulation. As a result, we optimized the Si_3N_4 nanopillars for designing efficient MIR metalens.

The optimized Si_3N_4 nanopillar meta-atoms exhibit high transmittance and minimal optical losses, as indicated by the average transmission efficiency of 97.44% shown in figure 3(a), which is essential for practical applications [37]. These nanopillars' transmittance and reflectance spectra, with a height of 20 μm and a radius of 470 nm, demonstrate their effectiveness in the MIR region. The inset schematic illustrates the Si_3N_4 nanopillars on a SiO_2 substrate, with geometrical parameters varied during optimization: radius (200–1260 nm), height (8–20 μm), and periodicity (3000 nm). These parameters significantly influence the optical response of the meta-atoms. Figure S2 of the supplementary information shows that the designed nanopillar maintains a high transmittance efficiency with a varying radius and height in a broad range of MIR from 8 to 12 μm. This optimal performance demonstrates that these

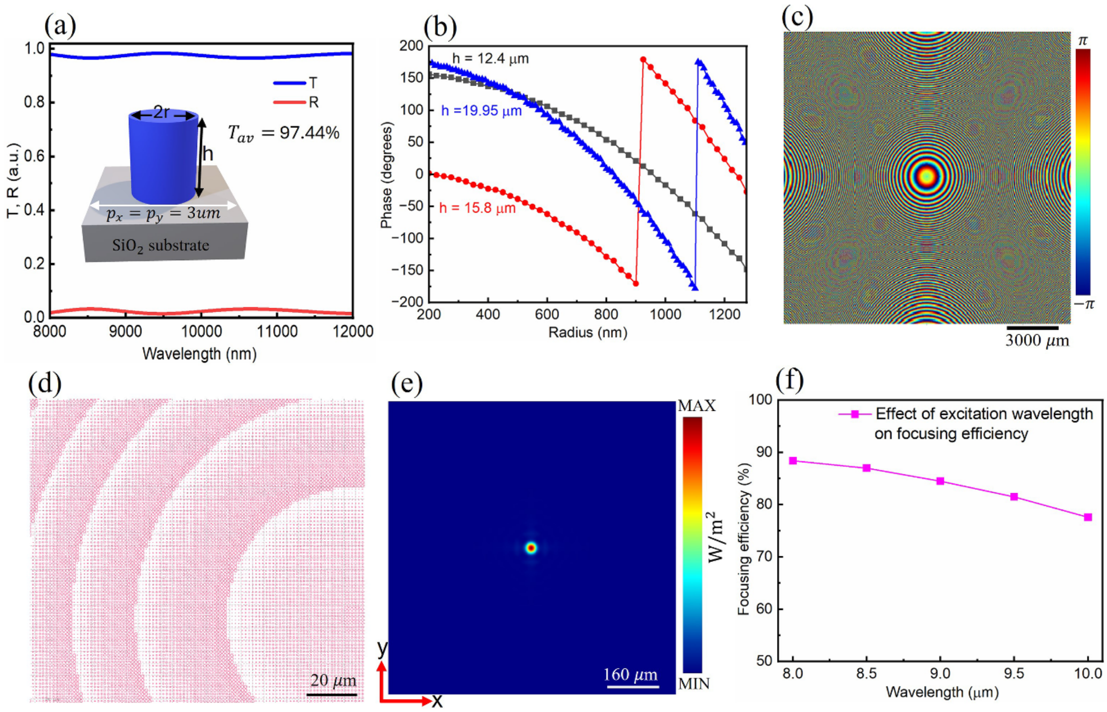

Figure 3. Optimization of Si_3N_4 nanopillars for MIR metalens. (a) The transmittance and reflectance of Si_3N_4 nanopillar meta-atoms with a height of 20 μm and a radius of $r = 470$ nm. The calculated average transmission efficiency is 97.44%. The inset shows a schematic illustration of the Si_3N_4 nanopillars on a SiO_2 substrate, with a radius ranging from $r = 200$ to 1260 nm, a height of $h = 8 - 20$ μm, and a periodicity of $P = 3000$ nm. (b) The phase shift induced by the Si_3N_4 metasurfaces at an excitation wavelength of 8 μm. (c) The phase target generated for the metalens at an excitation wavelength of 8 μm using equation (1) at a focal length of 1.5 mm. The corresponding (d) generated GDS shows the meta-atoms arrangements, and (e) the focusing ability at NA of 0.707. The designed metalens' root mean square error (RMSE) is 0.004, demonstrating high precision in matching the target wavefront and optimized accuracy in focal intensity distribution. (f) The focusing efficiency of metalenses varies with design wavelength.

nanopillars are ideal for fabricating efficient metalenses with high-quality focusing capabilities. Achieving a full 2π phase coverage is crucial for accurate wavefront shaping in metalens design. Figure 3(b) depicts the phase shift induced by the Si_3N_4 meta-atoms at an excitation wavelength of 8 μm. The ability of the nanopillars to provide the required phase shifts while maintaining high transmittance demonstrates the effectiveness of the optimization process [38]. The phase profile generated for the metalens, calculated using equation (1) at a focal length of 1.5 mm, is shown in figure 3(c). The curved wavefront indicates the lensing effect, confirming the metalens's capability to focus incident light.

The corresponding focusing ability is illustrated in figure 3(e), demonstrating effective light concentration at the focal point. Subsequently, the optimized Si_3N_4-based metalens at an excitation wavelength of 8 μm resulted in high focusing efficiencies of 91.5%, 88.4%, and 68.7% for NAs of 0.447, 0.707, and 0.98, respectively. These efficiencies are notably high when compared to other MIR metalenses [32, 39]. Moreover, our design achieves higher efficiencies and maintains them across a range of wavelengths (see figure 3(f)), highlighting its versatility and robustness for broad applications.

To the best of our knowledge, our all-pass Si_3N_4 metasurface filter achieved the highest focusing efficiency for high-NA metalens across a broad range of wavelengths. As a result, table 1 summarizes the performance parameters of reported metalenses and compares them to our results. The key novelty we introduce is that our metalens, with a large (NA $>$ 0.707), have demonstrated a focusing efficiency exceeding 85%. Additionally, the Si_3N_4 nanopillars used in our design secured nearly ideal transmission efficiency in the NIR regime of 1000–1600 nm, and 8–12 μm, enabling high-efficiency metalenses. This exceptional performance is attributed to the optimized geometry of the nanopillars, which allows for precise phase control of transmitted light while minimizing scattering and absorption losses. The low optical absorption of Si_3N_4 in these wavelength ranges contributes to efficient light manipulation, resulting in improved focusing efficiency.

Table 1. Focusing efficiency of metalenses at different numerical apertures (NAs). It also compares the transmission efficiency (T. eff) of the designed metasurface at given wavelengths.

			Metalens performance		
Material/substrate	Wavelength	T. eff.	NA	Focusing eff.	Reference
a-Si/fused silica	715 nm	88%	0.99	35%	[19]
SiO_2/BK7	520 nm	—	0.12	82%	[25]
Si_3N_4/SiO_2	1120 nm	81.9%	0.25	77.7%	[40]
Si_3N_4 nanoposts/SiO_2	430–780 nm	≈47%	0.086	36%–55%	[41]
Si_3N_4 nanoposts/SiO_2	550 nm	—	0.2	89%	[42]
TiO_2/SiO_2	500–1000 nm	75.78	—	54.24%	[43]
Si_3N_4/SiO_2	1100 nm	98.34	0.707	85.7%	our work
Si_3N_4/glass	450–1400 nm	77%	0.41	$>70\%$	[44]
PbTe/CaF_2	5.11–5.405 μm	80%	0.71	75%	[32]
Si_3N_4/SiO_2	8 μm	97.44	0.707	88.4%	our work
Si_3N_4/SiO_2	8 μm	97.44	0.98	68.7%	our work

3.4. Generation of optical vortex beams

Combining the functions of a lens and a spiral phase plate into a single metasurface allows us to generate focusing optical vortex (FOV) beams, with the resulting phase distribution profiles defined as follows [45]:

$$\varphi_m(r,\phi) = -\frac{2\pi}{\lambda}\left(\sqrt{r^2+f^2}-f\right)+l\cdot\phi \tag{2}$$

where $r^2 = x^2+y^2$, ϕ is the polar angle, and l represents the topological charge. This is essential for generating an optical vortex beam as it defines the necessary phase distribution that combines the functionalities of a metalens and a spiral phase plate into a single metasurface. This phase profile integrates a linear phase gradient for focusing light with an azimuthal phase variation to impart the helical structure required for vortex beam formation. The focal length, f, ensures the light is focused at a specific point while simultaneously forming the vortex beam. Using equation (2), the spatial distribution of the phase profiles for the metasurfaces was constructed. As a result, the spatial phase distribution was converted into Si_3N_4 nanopillars with varying radii R (see figures S4(a) and (b) in the supplementary information). This conversion is evident in figures 4(a)–(c), where the FOVs generated with topological charges of $l=1$ to $l=5$ at a focal length of 20 μm are demonstrated. Thus, show the intensity profiles of the FOVs on the focal plane ($z=20$ μm) with a (NA = 0.92). Doughnut-shaped focal spots are generated with high efficiencies, showcasing the capability of Si_3N_4 nanopillars to effectively manipulate the incident light phase.

To model the generated FOV beams, we used a simplified theoretical equation for the intensity profile of a vortex beam carrying OAM with topological charge l [21]:

$$I(r,z,l) = \frac{2PV}{\pi\, l!\, w^2(z)}\left(\frac{2r^2}{w^2(z)}\right)^l \exp\left(-\frac{2r^2}{w^2(z)}\right) \tag{3}$$

where $I(r,z,l)$ represents the intensity at a radial position r and propagation distance z; PV is the peak power of the vortex beam; $w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}$ is the beam waist at position z, with w_0 being the beam waist at focus; $z_R = \frac{\pi w_0^2}{\lambda}$ is the Rayleigh range, where λ is the wavelength of 1100 nm; and $l!$ denotes the factorial of the topological charge l. The term $\left(\frac{2r^2}{w^2(z)}\right)^l$ reflects the radial dependence introduced by the OAM, while the exponential term defines the Gaussian envelope of the beam. This analysis shows that the topological charge l intrinsically modifies the intensity profile. For $l>0$, a central null emerges in the intensity distribution, forming a ring-shaped pattern that expands with increasing l, aligning well with our simulated data. This behavior arises from the vortex beam's helical phase structure, where the phase twists around the propagation axis, leading to destructive interference at the center. The alignment of theoretical predictions with simulated data for $l=1,3$, and 5 confirms that the Si_3N_4 nanopillars effectively generate beams with precise OAM characteristics. For example, for the FOV at $l=5$, the focusing efficiency of the focal spot reaches 90.7%, which is significantly higher than that achieved with silicon metasurfaces (≈70%) [45]. The enhanced purity of the vortex beam can be attributed to the near-perfect transmission properties of the Si_3N_4 meta-components (see figures S2 and S4(b)

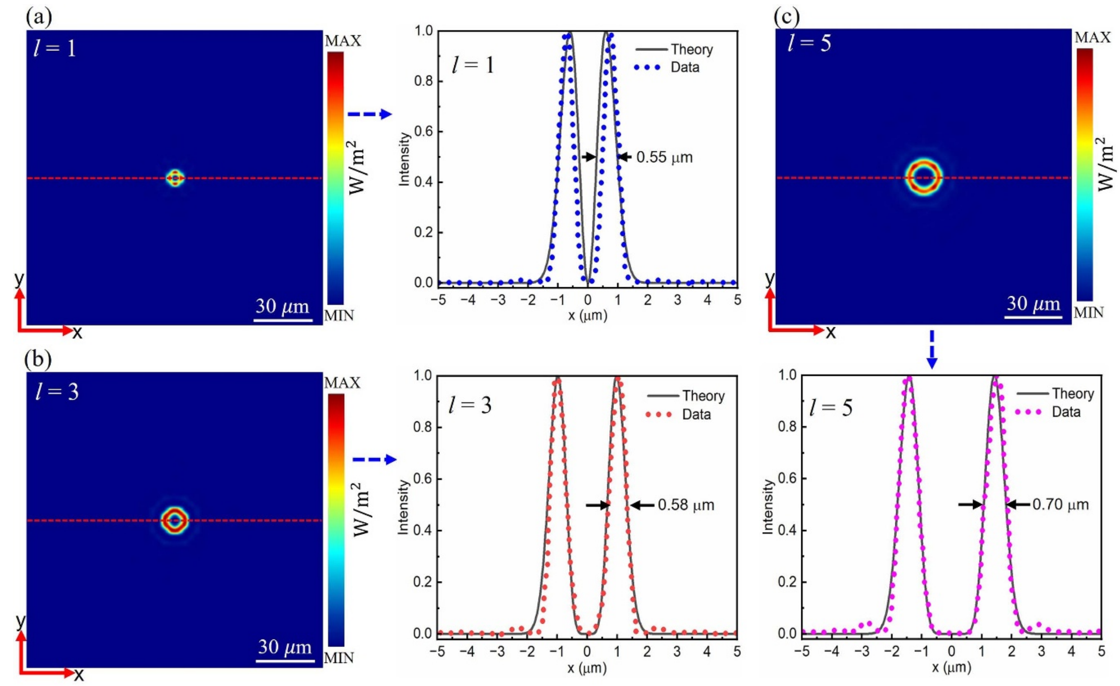

Figure 4. FOV generation at different l values. Focal spots and horizontal cross-sections of FOVs at the focal plane ($z = 20\ \mu$(m) with a numerical aperture (NA) of 0.92, for topological charges: (a) $l = 1$, (b) $l = 3$, and (c) $l = 5$. The FWHM of the doughnut-shaped focal spots' horizontal cuts (the dotted lines) are 0.55 μm, 0.58 μm, and 0.70 μm, respectively. The focusing efficiencies for the corresponding negative values of the topological charges (l) are also the same as those for the positive values, indicating symmetry in the system. The simulation data are consistent with the theoretical model fitted using equation (3), validating our design approach. Right-handed circular polarization (RCP) of the Gaussian beam at 1100 nm was used to generate the FOVs.

in the supplementary information). This results in well-defined optical vortices with minimal aberrations, even at higher topological charges. The ability to generate high-quality FOVs with varying l values is crucial for applications requiring precise control over the angular momentum of light. As the topological charge increases, the diameter of the doughnut-shaped focal spot also increases, which is consistent with theoretical predictions for optical vortices [46]. This behavior allows for tailoring the spatial intensity distribution of the focal spot to suit specific application needs. The high focusing efficiency and beam purity demonstrated here highlight the potential of Si_3N_4 metasurfaces in advanced photonic systems, such as optical trapping and quantum information processing.

Furthermore, the demonstrated vortex beams at different NAs have achieved high focusing efficiencies, representing a significant advancement over previous designs [47]. As illustrated in figures 5(a), (b) and 4(c), the generated FOVs exhibit well-defined intensity distributions at the focal plane, consistent with the metalenses' performance (see figures 2(b)–(e)). The higher NA enhances the spatial confinement of light, resulting in tighter focal spots within the characteristic donut-shaped intensity profile of vortex beams. This effect is attributed to the increased ability of high-NA metalens to collect and focus light from a wider range of angles, thus improving the resolution and intensity distribution of the focal spot. The tight focusing of vortex beams is crucial for applications requiring high spatial resolution and precision, such as optical tweezers [48].

Additionally, the Si_3N_4 nanopillars employed in the design exhibit polarization insensitivity, leading to consistent focusing efficiencies of the FOV for various input polarization states, including left-handed circularly polarized (LCP), TE, and right-handed circularly polarized (RCP) light (see figures 5(c), (d) and 4(c), respectively). Unlike approaches that rely on splitting energy between co- and cross-polarized outputs, our design achieves polarization insensitivity through the symmetric geometry of the nanopillars, which do not preferentially interact with any specific polarization state [49]. Such robustness against polarization variations is advantageous for practical implementations where the polarization state of the incident light may not be well-controlled. The high efficiency and polarization-insensitive generation of FOVs have significant implications for practical applications. For instance, in quantum information processing, vortex beams carrying optical angular momentum states can encode information in higher dimensions, enhancing the capacity and security of quantum communication systems [50].

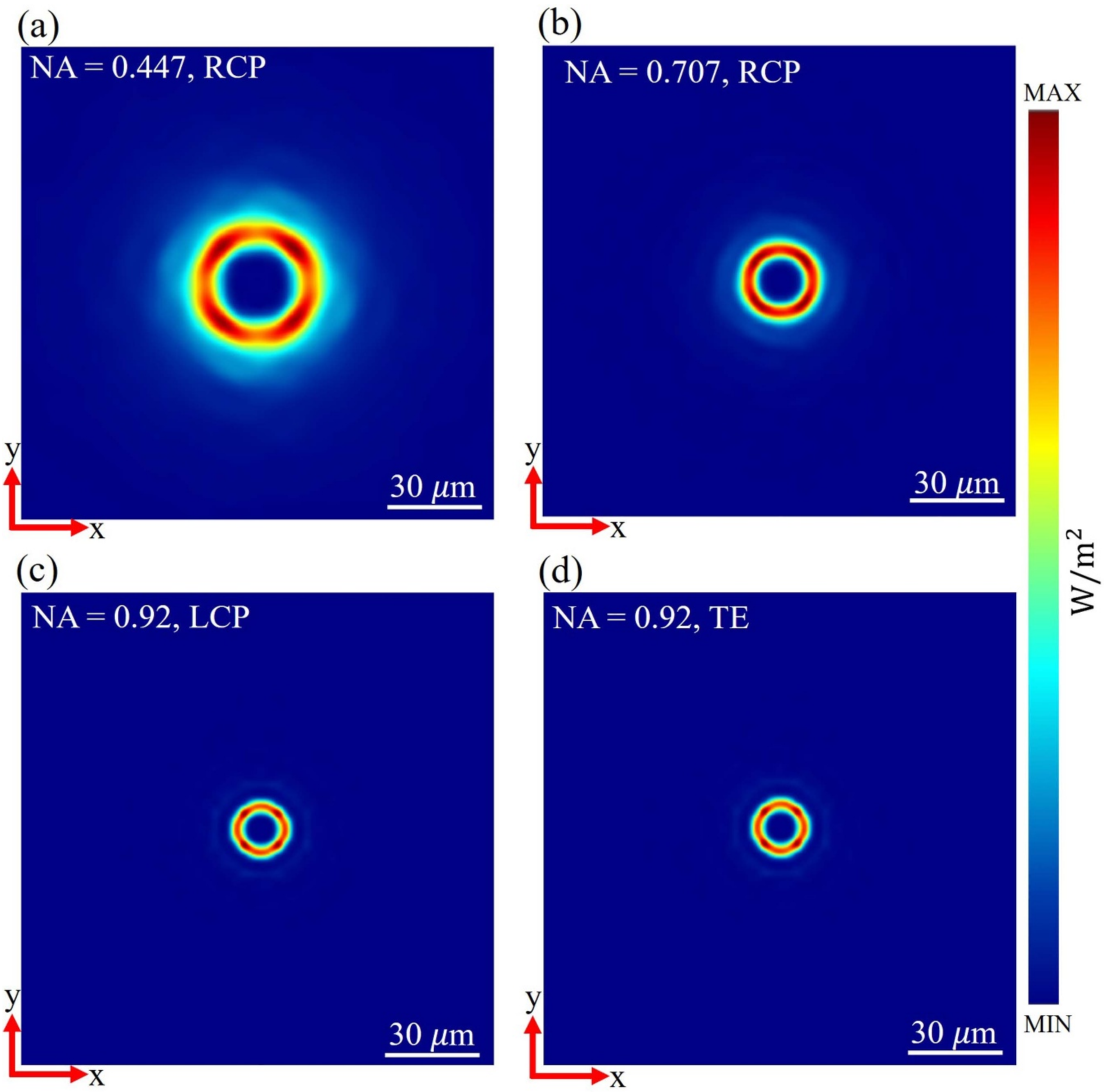

Figure 5. Generation of FOVs at different NAs and polarizations. (a), (b) FOVs generated at the focal point with numerical apertures of (a) 0.447 and (b) 0.707, demonstrating that higher NAs lead to tighter focusing and more pronounced vortex intensity distributions. (c), (d) Demonstration of polarization insensitivity of the FOVs for normal incidence of (c) LCP and (d) linear polarization with polarization angle 90°, compared with RCP shown in figure 4(c). For (c) and (d), the NA and topological charge (l) are fixed at 0.92 and 5, respectively, illustrating that the FOVs maintain their characteristic helical phase fronts and intensity profiles regardless of the polarization state of the incident light at 1100 nm.

Furthermore, as shown in Supplementary Note 1 and figures S5–S6 of the supplementary note, we also demonstrate OAM generation without a metalens, where the phase implements the canonical helical structure $e^{i\ell\phi}$ [11, 51]. These results indicate that incorporating a metalens into the optical-vortex design is crucial for enhancing spatial confinement (see figure 5). Building on this, the FOV profile adds a spherical phase curvature to the helical term, driving the wavefront toward a focal plane and concentrating the optical energy into a sharply defined annulus. Numerical results in figures S5 and S6 in the supplementary information, confirm that both beams maintain an OAM purity exceeding 90% for $\ell = 5$, indicating that nearly all optical energy resides in the targeted mode. However, high modal purity alone does not set the radial profile. With the same topological charge $\ell = 5$, the FOV's focusing phase confines energy into a compact annulus, whereas the canonical OAM beam excites additional radial components and remains less confined. To further substantiate this effect and demonstrate the metasurface's ability to sustain purity at higher topological charge, we computed vortex beams for $\ell = 10$. As shown in figures S7(a)–(c) in the supplementary information, the generated beams preserve purity above 90% while the FOV maintains superior spatial confinement. These results underscore the scalability and robustness of the approach for large-scale photonic applications.

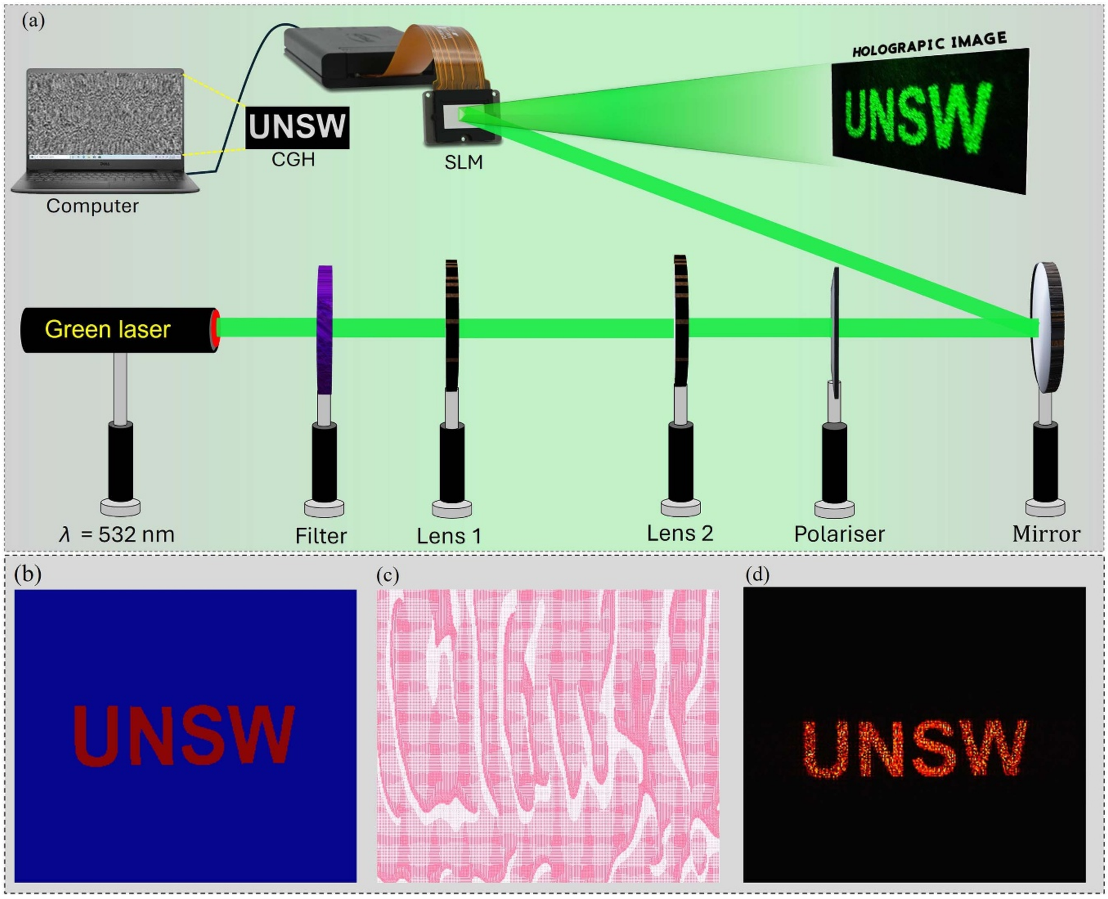

Figure 6. (a) Schematics of the experimental setup for generating reflective holographic images of the CGH of 'UNSW' using a spatial light modulator (SLM). The phase of the generated holographic image (top-right panel) matches with its corresponding CGH. Numerical calculations using reflective TiO_2 metasurface (b) the target phase pattern, (c) generated GDS, and (d) the reconstructed reflective holographic image. The wavelength of the incident light was set to 532 nm for both conditions.

3.5. Transmissive holography

Having established Si_3N_4 metasurfaces as a versatile platform for metalenses and optical vortex beam generation, we next explored their application in transmissive holographic imaging. However, before proceeding to investigate the transmissive holographic imaging using Si_3N_4, it is critical to validate that the phase maps computed using the modified Gerchberg–Saxton algorithm [52] can be accurately reproduced in an optical system. To this end, figure 6(a) illustrates a reflective-mode experimental setup, where a SLM displays the CGH of the phase patterns of the abbreviation UNSW. As can be seen clearly, the target phase and experimentally captured intensity profiles exhibited excellent agreement. Additionally, to further confirm the effectiveness of our simulation method in generating the transmissive holographic mode, we designed a reflective metasurface made of a TiO_2 metasurface with varying radius of the cylindrical pillars on a silver film at excitation wavelength of 532 nm. By varying the radii from 30 to 90 nm, we achieved a full 2π phase modulation. These nanopillars were arranged as shown in the figure 6(c) according to the phase of figure 6(b), producing a reflective hologram. Thus, a high resolution reflective holographic image resulted with an efficiency exceeding 80% (see figure 6(d)). Both the simulated and experimentally acquired reflective holographic images shown in figures 6(a) and (d) reveal comparable spatial resolutions, confirming the robustness of the phase-mapping algorithm employed. This high degree of agreement highlights the effectiveness of the numerical approach in accurately modeling realistic optical conditions.

Building upon the successful reflective hologram, we extended the validated CGH of 'UNSW' logo to design a transmissive hologram using Si_3N_4 metasurface. This choice is due to the high transmission efficiency of Si_3N_4 nanopillars, which is a crucial condition for efficient imaging [53]. Figure 7 illustrates the capability of Si_3N_4 metasurfaces to reconstruct holographic images at various incident wavelengths. As a result, in line with the reflective mode, the same phase pattern required to produce the desired image was used (see figure 7(a)). The resulting phase patterns were encoded onto the Si_3N_4 nanopillars

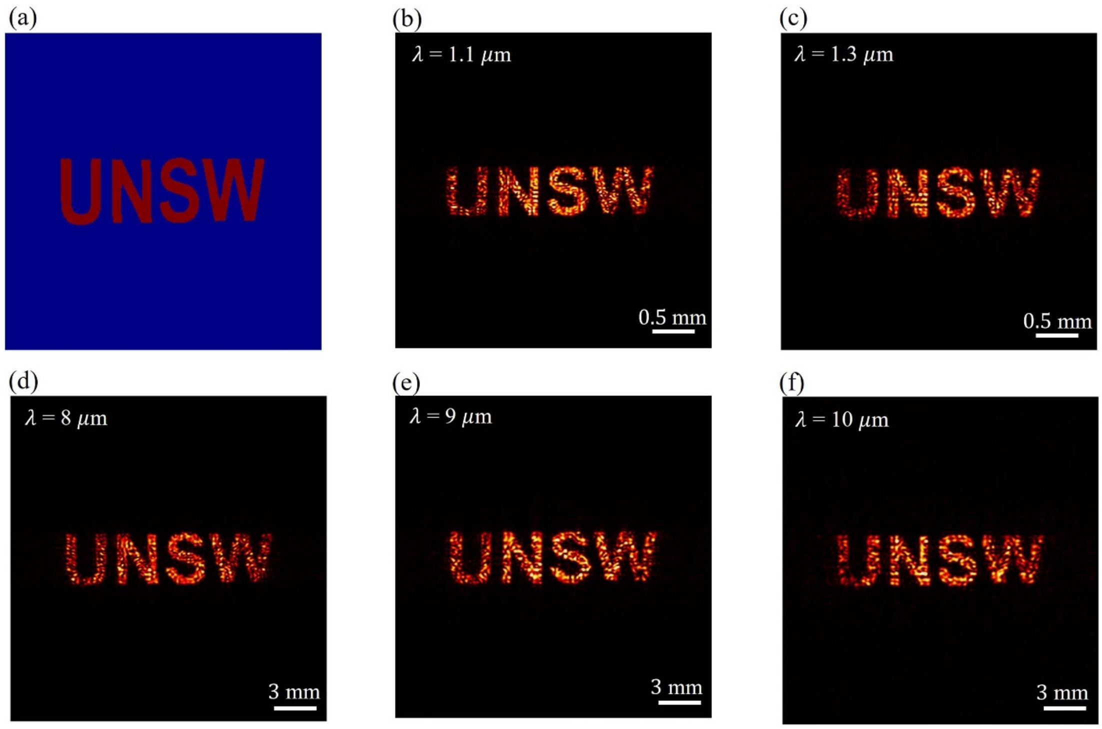

Figure 7. Transmissive Holographic imaging at different wavelengths. (a) The designed phase patterns for transmissive holographic imaging of 'UNSW' were implemented using Si_3N_4 metasurfaces designed for NIR and MIR. The transmitted beams form holographic images in the far field at excitation wavelengths of (b) 1.1 μm and (c) 1.3 μm in the NIR regime. In the MIR regime, holographic images are formed at wavelengths of (d) 8 μm, (e) 9 μm, and (f) 10 μm. For the MIR regime, the images were constructed over a larger area of 3×3 mm^2 and reconstructed in the far field at $z = 0.8$ cm.

using PlanOpSim software, enabling precise manipulation of the phase of transmitted light to reconstruct the target image. Figures 7(b) and (c) depict the far-field images at an excitation wavelength of 1100 nm and 1300 nm. The nearly consistent quality of the images across different wavelengths demonstrates the metasurface's ability to control light propagation effectively. The implementation of Si_3N_4 metasurfaces significantly extends the capabilities of imaging into the MIR spectrum, specifically at operating wavelengths ranging from 8 μm to 10 μm, as depicted in figures 7(d)–(f). The reconstructed images demonstrate high efficiency and consistent resolution across this broadband MIR range. At an excitation wavelength of 8 μm, the hologram efficiency reaches up to 80%, which is higher than that of the previously reported hologram image using silicon nanodisks [15]. Remarkably, this efficiency remains nearly constant across the MIR spectrum, with values of 77.2% at 9 μm and 76.0% at 10 μm. The demonstrated efficiency is higher than that of silicon nitride (SiN) nanopillars, which have a 73% diffraction efficiency due to their 92% transmittance efficiency of the metasurface [54].

This consistency indicates the robustness of Si_3N_4 metasurfaces filters for broadband applications and highlights their potential for versatile optical devices such as multispectral imaging. The high transmission efficiency enhances the brightness and clarity of the holographic images, which is essential for practical optical applications [29]. Additionally, the nearly all-pass filter behavior allows for efficient phase modulation without compromising the amplitude of the transmitted light, enabling high-fidelity hologram reconstruction [7].

In line with the design of hologram imaging of 'UNSW' at different excitation wavelengths shown in figure 7, the UNSW logo was also reconstructed at multiple propagation distances. As shown in figures S8(b)–(d) in the supplementary information, the holographic images were successfully reconstructed in the far-field at an excitation wavelength of 1100 nm. The nearly consistent resolution of the images across different distances demonstrates the metasurface's ability to control light propagation effectively. This spatial flexibility is crucial for applications like AR and optical data storage, where holographic elements must function effectively over varying distances [55, 56]. As a result, our results align with the work of Zheng *et al*, who achieved metasurface hologram efficiencies of up to 80% in the visible spectrum [55]. Our work thus paves the way for developing efficient holographic devices that operate effectively across a wide spectral range, addressing a critical need in advanced optical technologies.

Lastly, it is noteworthy to explain that the practical feasibility of our design can be fabricated using a standard nanofabrication facility. The fabrication of a Si_3N_4 metasurface using electron-beam lithography (EBL) begins with the deposition of a high-quality Si_3N_4 film on a SiO_2 substrate through plasma-enhanced chemical vapor deposition [57]. A thin metallic layer, such as chromium, is first evaporated to form a hard mask, followed by the spin-coating of an electron-beam resist on top. Then, EBL is used to define the nanoscale patterns, which are then developed and transferred to the mask layer using a standard lift-off and etching process. Finally, these patterns are etched into the Si_3N_4 film using inductively coupled plasma etching with carefully optimized gas mixtures to achieve smooth sidewalls and highly anisotropic profiles. This top-down approach makes it possible to realize high-aspect-ratio Si_3N_4 nanostructures that manipulate light effectively across a broad range of spectrum.

4. Conclusion

In this study, we design and analyze metalens, optical vortex beams, and holographic imaging systems using Si_3N_4 metasurfaces composed of nanopillars. We investigated the performance of these devices in the NIR regime at 1100 nm and the MIR regime at 8000 nm. Our findings demonstrate that Si_3N_4-based metalenses achieve high focusing efficiencies across various NAs. Notably, at an NA of 0.707, the metalenses maintain a focusing efficiency greater than 85%, which is applicable for high-resolution imaging. In addition to metalenses, we successfully generated high-quality FOV beams by integrating the functions of a lens and a spiral phase plate into Si_3N_4 metasurfaces. At an operating wavelength of 1100 nm and a topological charge of $l = 5$, the FOV beams achieved a focusing efficiency of 90.7%. The near-perfect transmission properties of the Si_3N_4 nanopillars contribute to the vortex beams' enhanced purity and minimal aberrations, even at higher topological charges. This capability is crucial for applications requiring precise control over the angular momentum of light, such as optical trapping and quantum information processing. Furthermore, after experimentally verifying the accuracy of the CGH phase using the SLM, transmissive holographic imaging was demonstrated using Si_3N_4 metasurface at excitation wavelengths of NIR and MIR. The reconstructed holographic images have a good resolution, with efficiencies reaching up to 80%. These results underscore the potential of Si_3N_4 metasurfaces in advancing photonic systems by providing high efficiency and versatility across multiple applications. The study firmly establishes that Si_3N_4 metasurfaces can achieve superior efficiencies in both NIR and MIR regimes, opening avenues for their integration into advanced optical devices.

Data availability statement

All data that support the findings of this study are included within the article and the supplementary files.

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Conflict of interest

The authors declare no conflict of interest.

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